Vincent S. Pronel

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Education

Bachelor of Science in Computer Science **Expected 2027**
Kean University Union, NJ

- **Relevant Coursework**: Software Engineering*, Analysis of Algorithms*, Computer Architecture*, Object Oriented Analysis & Design, Database Management Systems, Applied Statistics, Matrix & Linear Algebra **in progress, Fall 2026*
- **Leadership & Honors**: Web Developer, WiCyS (Women in Cyber Security) Kean Chapter E-Board | Presenter, 2026 CRA UR2PhD Research Showcase, New Orleans | GMiS Research Posters Competition | Dean's List, Spring 2024 and Fall 2025

Technical Skills

- **Languages**: TypeScript, JavaScript, Java, Python, SQL, PHP, HTML/CSS
- **Frameworks & APIs**: React, Next.js, Node.js, REST APIs, Java Debug Interface (JDI), JavaParser, Tailwind CSS
- **AI, Data & Analytics**: OpenAI and Google Gemini APIs, supervised fine-tuning, prompt design, model evaluation, data visualization, correlation analysis
- **Cloud & DevOps**: Cloudflare Workers, D1, R2, Pages, Vercel CI/CD, Supabase, PostgreSQL, MySQL, Railway
- **Tools**: Git/GitHub, Linux, MySQL Workbench, Figma

Professional Experience

Independent Web Developer Jan. 2026 – Present
Freelance Full-Stack Web Development *Sayreville, NJ*

- Founded and run a freelance practice delivering 5 production websites for 5 paying business clients and building the WiCyS Kean Chapter site; independently own requirements, architecture, implementation, deployment, and maintenance.
- Build with **Next.js**, **TypeScript**, React, PostgreSQL, Supabase, and **Cloudflare Workers**; delivered live payment functionality for 2 clients and admin tools used by approximately 2 staff members per business.
- Selected work: daysicecream.com (4,000+ monthly visitors) | trainingaio.com (booking platform, delivered in 2 weeks) | tedishairstudio.com | deafsharkcoffee.com | adzio.io | wicyskean.me.

Computer Science Tutor (Code Samurai) Jan. 2026 – Present
Kean University *Union, NJ*

- Selected by CS faculty to provide one-on-one tutoring in Database Management Systems, coaching undergraduates through relational schema design, SQL queries, and step-by-step debugging.
- Debug students' own code and queries with them rather than supplying answers, tracing logic step by step so they can reproduce the reasoning independently on later assignments.

Data Visualization & Analytics Intern July 2026
Group Summer Scholars Research Program, Kean University *Union, NJ*

- Lectured daily and mentored 5 high school students under *Dr. Ching-yu Huang*, teaching relational querying and interactive charting with SQL, PHP, MySQL, JavaScript, and Google Charts.
- Independently developed 8 hands-on labs supporting the faculty-led curriculum; all 5 students completed database-driven websites and presented individual research posters.

Research Projects

Java Execution Visualizer | *CRA UR2PhD-REU Fellowship* June 2026 – Present
Undergraduate Research Fellow, Advisor: Dr. Yan Ma | Competitively selected, $8,000 funded, 400 hours *Kean University*

- Independently built a Next.js/TypeScript Java memory visualizer with 4 guided program simulations and 26 execution states, animating stack frames, heap objects, arrays, aliasing, method calls, and reference changes line by line.
- Engineered an optional adapter for a JDI-based Java tracer that normalizes raw JVM stack and heap data into interactive states while enforcing 15-second, 4 MiB, 20,000-character, and 300-step resource limits.
- Built an IRB-approved 2-condition study platform targeting 40 completed sessions, with balanced assignment, idempotent event logging, hashed session tokens, abuse controls, and paired pre/post analytics in Supabase/PostgreSQL.

Jobs vs. Wages | *NSF IUSE Scholars Academy* May 2026 – Present
NSF IUSE Research Scholar, Advisor: Dr. Pauline Wang | Competitively selected, $1,400 funded, 10 weeks *Kean University*

- Built a MySQL data mart and interactive dashboard over 33,000 records from the US ZIP Codes Database, writing the population-weighted aggregation for all 50 states, Washington, D.C., and Puerto Rico; presented the results at the 2026 GMiS conference in Albuquerque.

AI Debate Coach Evaluation | *CRA UR2PhD Research Program* Feb. 2026 – Apr. 2026
Undergraduate Researcher, Advisor: Dr. Yan Ma *Kean University*

- Led development of a Node.js/Express research platform comparing 3 AI debate configurations, implementing SSE response streaming, Neon/PostgreSQL persistence, retry handling, structured surveys, and an authenticated analytics and export dashboard.
- Built a 3-round study workflow capturing five-exchange transcripts, 24 Likert ratings, comparative rankings, qualitative feedback, and automated evaluation results for each participant.